Prabhavit Gupta

AI Engineer | Agent Frameworks, Trace Intelligence, MCP & Automated Evaluation

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Executive Summary

- **Production Agent Architecture & Framework Engineering:** AI engineer with 3 years commercial experience architecting, benchmarking, and shipping autonomous agent systems, tool loops, and agentic primitives. Proficient in designing stateful multi-agent execution graphs, short/long-term memory backends, and modular tool interfaces in TypeScript and Python.
- **Trace Intelligence & Automated Evaluation (Auto-Evals):** Deep hands-on experience instrumenting distributed agent trace observability (Langfuse, OpenTelemetry) and transforming execution traces into actionable insights. Built automated evaluation pipelines (synthetic benchmark generation, RAGAS, tool-call precision metrics) to detect reasoning drift and continuously fine-tune agent behavior.
- **Standardized Protocols & Systems Engineering:** Specialist in Model Context Protocol (MCP Server & Client SDKs), exposing secure database tools and APIs to autonomous agents. Strong production foundation across Docker, Redis, PostgreSQL (pgvector), and automated CI/CD regression suites.

Technical Skills & Core Competencies

- **Agent Frameworks & Primitives:** Autonomous Agents, Multi-Agent Orchestration, Tool Calling, Memory Management (Working / Episodic / Semantic Memory), State Persistence, Human-in-the-Loop, Context Pruning.
- **Trace Intelligence & Observability:** Distributed Execution Tracing, Trace Mining, Langfuse, OpenTelemetry, Latency Profiling, Token Cost Accounting, Decision Path Auditing.
- **Evaluation & Benchmarking:** Auto-Evals, Synthetic Dataset Generation, RAGAS Benchmarking, Regression Detection Suites, Tool Call Accuracy Scoring, Model Routing & Rollouts.
- **Protocols & Data Backends:** Model Context Protocol (MCP), REST / WebSocket APIs, PostgreSQL (pgvector), Redis (Caching & Semantic Queues), Pydantic / Zod Structured Output Schemas.
- **Languages & Tooling:** TypeScript / Node.js, Python (AsyncIO, FastAPI, PyTest), Docker, Microservices, Git, GitHub Actions CI/CD.

Professional Experience

AI Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote Global)

- Designed, benchmarked, and shipped production-grade agentic platforms and autonomous tool-calling systems for European tech organizations.
- Developed and open-sourced custom Model Context Protocol (MCP) servers and clients, standardizing agent tool interactions against relational schemas and external web APIs.
- Built trace intelligence pipelines leveraging Langfuse and OpenTelemetry, parsing multi-turn agent logs to detect execution failures, cyclical loops, and unnecessary token burn.
- Established automated evaluation harnesses to benchmark agent primitives (memory retrieval, function-calling precision, reasoning steps) against curated golden datasets before production rollouts.
- Optimized memory retention algorithms by pairing semantic embedding caching (Redis) with dynamic context compression, cutting median response latency by 35% and reducing API cost.
- Engineered automated CI/CD regression gates via GitHub Actions, validating that framework updates did not degrade task success metrics across complex agent execution paths.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Built asynchronous backend pipelines in Python and TypeScript for high-volume automated data processing, semantic parsing, and classification.
- Implemented vector search and retrieval pipelines using PostgreSQL (pgvector) and Redis, ensuring reliable sub-second knowledge access for downstream LLM agents.
- Designed comprehensive automated testing suites with PyTest and Vitest, ensuring defensive input handling and preventing regression in distributed microservices.
- Collaborated with product engineers to translate abstract developer workflows into modular, reusable architectural patterns.

Key Engineering Projects

- **Autonomous Tool-Using Agent Platform (TypeScript + Python + LangGraph + MCP):** Multi-agent collaborative execution environment coordinating verified tool executions via Model Context Protocol, featuring state

persistence, memory checkpointing, and Langfuse tracing.

- **High-Throughput Financial Intelligence & Extraction Engine (FastAPI + pgvector):** Retrieval and structured data extraction system with automated Pydantic schema validation, cross-encoder re-ranking, and regression evaluation harnesses.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous foundation in formal logic, statistical reasoning, decision theory, and complex systems analysis.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency).
- **Work Authorization & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted legal right to live and work across the EU without sponsorship; available immediately for remote engagement worldwide.